



Eigenmind - Cognitive Maps

From Implementation to Mathematics and Semantic Interpretation

Merlin Intelligence

May 20, 2026

Contents

1	Purpose and Reading Guide	3
2	Document Ingestion	3
2.1	Step 1 — Text Extraction (<code>document_loaders.py</code>)	3
2.1.1	Main steps	3
2.2	Step 2 — Chunking (<code>chunking.py</code>)	4
2.2.1	Main steps	4
2.3	Step 3 — Embedding (<code>embeddings.py</code>)	6
2.3.1	Main steps	6
2.4	Step 4 — Indexing (<code>vectordb/store.py</code>)	7
2.4.1	Main steps	7
3	Vector Similarity	9
3.1	Step 1 — The Cosine Similarity Matrix (<code>singular.py</code>)	9
3.1.1	Main steps	9
3.1.2	What it produces	9
3.2	Step 2 — Top- <i>k</i> Retrieval (<code>store.similarity_search</code>)	10
3.2.1	Main steps	10
3.2.2	Two retrieval regimes	11
4	Graph Exploration	12
4.1	Step 0 — Building the Subgraph (<code>exploration.py</code>)	12
4.1.1	What BFS means here	12
4.1.2	Main steps	12
4.1.3	What it produces	13
4.2	Strategy 1 — Thematic Diversity (<code>singular.py</code>)	13
4.2.1	Main steps	13
4.3	Strategy 2 — Hinge Connectors (<code>connectivity.py</code>)	14
4.3.1	Main steps	14
4.4	Strategy 3 — Frontier Themes (<code>theta.py</code>)	15
4.4.1	Main steps	15
4.5	The Pipeline in One Picture	18
4.6	Practical Scope, Assumptions, and Failure Modes	18

4.6.1	Intended operating regime	18
4.6.2	Formal assumptions	19
4.6.3	Known failure modes	19
4.6.4	Empirical validation (future work)	19
4.7	Aggregation — The <code>SimilarityGraph</code> Class	19
4.7.1	Main steps	19
4.7.2	The three-label output	19
4.7.3	What multi-label chunks reveal	20
4.7.4	Design principle: named chunks, not latent vectors	20
4.8	What Is Formal, What Is Heuristic, What Is Empirical	20
4.9	Summary	22

1 Purpose and Reading Guide

This memo inverts the usual presentation order. Instead of starting from mathematical theory and descending to code, it starts from what is *actually implemented* in the Python modules :

- document ingestion
- vector similarity
- graph exploration
- ontology learning

Then, it asks: *why does this work?* and *what does it mean for a corpus of strategic documents?*

Each of the ranking strategies is presented in the same order:

1. **Main steps:** the key algorithmic operations, in mathematical notation.
2. **Mathematical grounding:** the theorem or formal result that justifies the construction.
3. **Semantic interpretation:** the strategic meaning of the retrieved chunks, illustrated with a concrete example.

Notation convention. The memo distinguishes three levels of claim throughout:

- **Formal theorem:** a statement with a proof (e.g. Cheeger inequality, strong duality).
- **Mathematical intuition:** a heuristic argument grounded in theory but not a formal guarantee in this specific context.
- **Semantic interpretation:** a heuristic mapping from a mathematical object to a strategic concept; suggestive rather than proven.

The [grey boxes](#) carry mathematical grounding; the [dark-bordered boxes](#) carry heuristic semantic interpretation with graph illustrations. Claims in the semantic boxes should be read as working hypotheses to be validated empirically, not as theorems.

2 Document Ingestion

The ingestion pipeline turns a heterogeneous corpus of files into a collection of embedded, addressable chunks stored in Qdrant. The implementation is organised in four sequential stages: text extraction (`document_loaders.py`), chunking (`chunking.py`), embedding (`embeddings.py`) and indexing (`vectoradb/store.py`). The orchestrating class `Ingestor` in `pipelines/ingest.py` threads these stages together and exposes two strategies: `run_chunknorris` for structure-aware PDF ingestion and `run_multi_format` for the general PDF/DOCX/PPTX/XLSX/TXT/MD pathway.

2.1 Step 1 — Text Extraction (`document_loaders.py`)

2.1.1 Main steps

1. **Format dispatch.** Given a file f with extension $\text{ext}(f)$, dispatch to a format-specific extractor:

$$T(f) = E_{\text{ext}(f)}(f) \in \Sigma^*,$$

where Σ^* is the set of finite Unicode strings. Handlers cover PDF (`pypdf`), DOCX (`python-docx`), PPTX (`python-pptx`), XLSX (`pandas`) and plain text.

2. **PDF heuristic + OCR fallback.** Let T_0 be the text extracted from the PDF text layer and n_p the page count. If $|T_0|/n_p < 20$ characters per page, the file is flagged as a

likely scan and OCR is invoked:

$$T(f) = \begin{cases} T_{\text{OCR}}(f) & \text{if } |T_0|/n_p < 20 \text{ and } |T_{\text{OCR}}| > |T_0|, \\ T_0 & \text{otherwise.} \end{cases}$$

OCR proceeds via `pdf2image` $\rightarrow$ `pytesseract`.

3. **Filtering.** Empty extractions are dropped before chunking; failures are logged per file and do not abort the run.

Mathematical Grounding

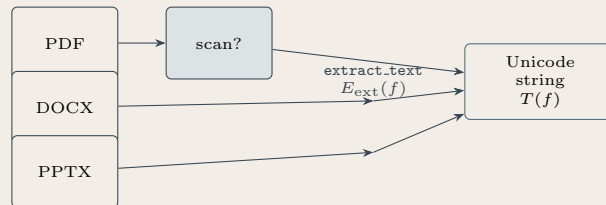
Mathematical framing: extraction as a noisy channel. Each file f carries an underlying authored text $T^*(f) \in \Sigma^*$; the extractor E implements a (lossy) decoding $\hat{T} = E(f) \approx T^*(f)$. For digitally produced documents, E is essentially exact; for scans, E collapses to optical character recognition, which is the maximum *a posteriori* reconstruction $\hat{T} = \arg \max_T p(T \mid I)$ over a sequence of page images $I = (I_1, \dots, I_{n_p})$.

Practical guarantees. The character-density test $|T_0|/n_p < 20$ is a simple but empirically reliable proxy for *textless pages* (an image-only PDF emits very few characters through the text layer). The OCR branch is invoked only when the cheap path fails, preserving ingestion throughput on well-formed corpora.

Limitations. The extractor inherits the failure modes of its underlying libraries: PDF text re-flow drops typographic order in multi-column layouts; PPTX shape iteration misses image alt-text; XLSX flattens spreadsheet semantics into a string serialisation. These artefacts propagate downstream as noise in the embedding space.

Semantic Interpretation (heuristic)

(Heuristic) The extractor preserves *surface* rather than *structure*.



The pipeline flattens every supported format into a single Unicode string, then forgets the format. This is deliberate: downstream stages only assume a stream of natural language text, never the existence of slides, tables or page numbers. The trade-off is that high-level document structure (sections, figures, footnotes) is partially lost; only the markdown-aware ChunkNorris branch recovers some of it for PDFs.

Analyst question: “What did the author actually write, regardless of how the file was packaged?”

2.2 Step 2 — Chunking (`chunking.py`)

2.2.1 Main steps

1. **Two strategies.** The implementation exposes a *lightweight* splitter (`langchain_chunker`) and a *structure-aware* pipeline (`chunknorris_pipeline`). Both produce a finite sequence of chunks $C = (c_1, \dots, c_m)$, $c_i \in \Sigma^*$, with $|c_i| \leq L$ (target chunk size, default $L = 300$ characters).

2. **Recursive character splitter (Langchain).** Given a separator priority list $S = (s_1, \dots, s_p) = (\backslash n \backslash n, \backslash n, ., \dots,)$, the splitter recursively partitions T along the highest-ranked separator that yields fragments of size $\leq L$. Overlap $o = 30$ characters is reinjected between consecutive chunks:

$$c_{i+1} \supseteq \text{tail}_o(c_i), \quad |c_i| \leq L, \quad |c_i \cap c_{i+1}| \approx o,$$

where $\text{tail}_o(c_i)$ denotes the last o characters of c_i .

3. **Markdown-aware splitter (ChunkNorris).** For PDFs, ChunkNorris first parses the file into a markdown intermediate $M = \pi_{\text{PDF}}(f)$ and then chunks M along its heading hierarchy. This preserves logical units (sections, lists, tables) at the price of higher latency and a heavier dependency stack.
4. **Empty-chunk filtering.** Whitespace-only chunks are dropped before embedding.

Mathematical Grounding

Mathematical framing: chunking as an overlapping cover. A chunker realises a map $\kappa : \Sigma^* \rightarrow (\Sigma^*)^m$ such that $T \subseteq \bigcup_i c_i$ (every character of T lies in at least one chunk). The overlap $o > 0$ ensures that for adjacent (c_i, c_{i+1}) the intersection $c_i \cap c_{i+1}$ is non-empty, which means that any sentence straddling a cut appears in at least one chunk in full. Formally, if L and o exceed the maximum sentence length in T , κ is a sentence-preserving cover.

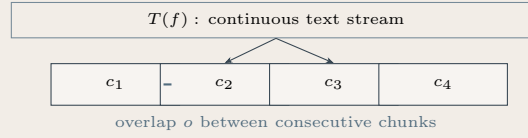
Recursive separator choice. The separator priority S encodes a coarse-to-fine semantic hierarchy: paragraph break $>$ line break $>$ sentence break $>$ word break (ordering by decreasing semantic strength). Greedy descent along S implements an approximate *minimum-disruption cut*: cut at the highest-ranked separator that respects the length budget L . This is equivalent to minimising a hierarchical cut cost $\sum_i w(s_{(i)})$ under a piecewise length constraint, where $w(s_k)$ assigns lower cost to higher-ranked separators.

ChunkNorris as a structural prior. The markdown intermediate M is a tree of (heading, body) pairs; ChunkNorris chunks along subtrees rather than character offsets. The induced chunks have higher *topical coherence* (each chunk covers one section), at the cost of unequal lengths.

Limitations. Both chunkers are length-driven, not topic-driven. A long sustained argument can be split mid-claim; a short heading-only section can produce a degenerate chunk. Adaptive, embedding-aware chunking is left to future work.

Semantic Interpretation (heuristic)

(Heuristic) Chunks are the readable atoms of the corpus.



Every downstream object—an embedding, a graph node, a singular pole, a retrieved passage—ultimately points back to one c_i that an analyst can read. This is a deliberate choice: the system never operates on opaque latent vectors alone, but always on *named, addressable text fragments*. The chunking strategy therefore determines the granularity of the analytical lens: too fine and arguments fragment, too coarse and contradictions disappear into the same chunk.

In strategic corpora, the empirical sweet spot is one chunk $\approx$ one paragraph or one sub-section—short enough for a single embedding to be coherent, long enough to carry a self-contained claim.

Analyst question: “What is the smallest unit of text I can still read, attribute and reason about?”

2.3 Step 3 — Embedding (embeddings.py)

2.3.1 Main steps

1. **Encoder.** The `EmbeddingModel` class wraps a `SentenceTransformer` backbone (default `all-MiniLM-L6-v2`, $d = 384$). Each chunk text c_i is mapped to a vector $\phi_i \in \mathbb{R}^d$:

$$\phi_i = f_\theta(c_i), \quad f_\theta : \Sigma^* \rightarrow \mathbb{R}^d.$$

2. **Device dispatch.** `detect_device()` returns the first available device in the priority order `cuda > mps > cpu`, and the model is loaded onto that accelerator. The context-manager protocol releases the model and clears the CUDA cache at the end of an ingestion run.
3. **Normalisation convention.** `SentenceTransformer`’s mean-pooled outputs for the default model are (approximately) ℓ_2 -unit: $\|\phi_i\|_2 \approx 1$. This property makes the downstream Gram-type similarity matrix (formally introduced in Section 3.1) approximate cosine similarities directly, without an explicit normalisation step.

Mathematical Grounding

Mathematical framing: chunks as points on the unit sphere. The encoder f_θ induces a (soft) map $\Sigma^* \rightarrow S^{d-1} \subset \mathbb{R}^d$. For a sentence-pair contrastively trained model (SBERT [3]), this map is optimised so that semantically equivalent sentences have small angular distance:

$$\cos(\phi(c), \phi(c')) \approx \mathbf{1}[c \equiv c'] + (\text{graded similarity for related pairs}),$$

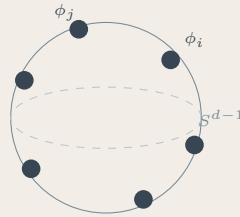
where $\mathbf{1}[\cdot]$ denotes the $\{0, 1\}$ -valued indicator function.

Dimension and capacity. With $d = 384$, the unit sphere S^{383} admits, by Johnson–Lindenstrauss-type concentration, an exponential number of ε -distinguishable directions. In practice, the encoder concentrates corpus chunks on a low-dimensional sub-manifold of S^{d-1} ; the embedded chunk cloud has effective rank much smaller than d , which is what makes the spectral analysis of the similarity matrix (Section 3) analytically meaningful.

Limitations. `all-MiniLM-L6-v2` is a general-purpose English encoder. Domain-specific corpora (legal, biomedical, multilingual strategic documents) typically benefit from a specialised encoder. *Embedding collapse*—distinct chunks mapped to nearby vectors—is the primary failure mode and propagates as spurious edges in W .

Semantic Interpretation (heuristic)

(Heuristic) The embedding turns text into geometry.



angular distance $\propto$ semantic distance

Each chunk becomes a point on a high-dimensional sphere, and the angle between two points measures how close their meanings are. This single geometric primitive—cosine of an angle—is what powers every subsequent stage: the similarity matrix W , the BFS retrieval, the Laplacian spectrum, the geodesic field, and the conflict graph.

The strategic implication is direct: the quality of the analytic output is upper-bounded by how faithfully f_θ aligns *textual semantics* with *angular geometry* on the corpus at hand. Choosing the encoder is therefore an editorial decision, not a technical one.

Analyst question: “Does the geometry I am about to reason on actually reflect the meanings I care about?”

2.4 Step 4 — Indexing (vectordb/store.py)

2.4.1 Main steps

1. **Collection bootstrap.** `ensure_collection(name, d)` creates a Qdrant collection with vector size d and cosine distance, or returns the existing dimension if the collection already exists.
2. **Point construction.** Each chunk yields a `PointStruct` $\pi_i = (\text{id}_i, \phi_i, p_i)$ where id_i is a

fresh UUID and the payload is

$$p_i = \{\text{filename}, \text{chunk_number}, \text{text}, \text{ingestion_date}\}.$$

The text itself is stored in the payload, so retrieval returns readable chunks without a side database.

3. **Deduplication.** `existing_filenames(collection)` scrolls the collection and returns the set of filenames already indexed; the ingester skips any file whose basename is already present.
4. **Batched upsert.** `batched_upsert` drains the point iterator into Qdrant in groups of `BATCH_SIZE = 100`, returning the total count. Failures inside a batch propagate; failures inside a single file are caught and logged so that one bad PDF does not abort the run.

Mathematical Grounding

Mathematical framing: HNSW as an approximate sphere index. Qdrant stores the embeddings under a Hierarchical Navigable Small World (HNSW) graph [1]: a layered proximity graph in which a greedy descent from a coarse layer down to the base layer returns approximate nearest neighbours in time $O(\log N)$ on average, for N stored vectors.

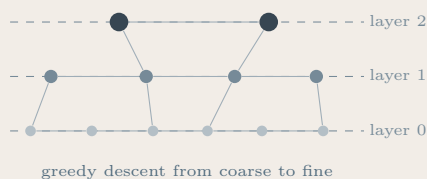
Cosine equivalence. For ℓ_2 -unit vectors ϕ_i , $\|\phi_i - \phi_j\|_2^2 = 2(1 - \phi_i^\top \phi_j)$, so Euclidean nearest-neighbour, maximum-inner-product and cosine similarity rankings coincide. Choosing `Distance.COSINE` in the collection config is therefore consistent with the implicit normalisation of SBERT-style encoders.

Payload as a side index. Storing `(filename, chunk_number, text)` in the payload means that the vector index and the document index live in the same store: a single `retrieve` call returns both the geometry and the readable content. `ingestion_date` enables time-windowed re-indexing and per-day deletion (`delete_for_date`).

Limitations. HNSW returns approximate, asymmetric neighbours (cf. Section 4 caveat). Deduplication is filename-level only: two distinct files containing identical chunks will both be indexed; semantic deduplication is left to downstream stages.

Semantic Interpretation (heuristic)

(Heuristic) The corpus becomes a navigable point cloud.



Once a file is ingested, every one of its chunks is a coordinate in the same shared space, reachable in $O(\log N)$ from any query. Newly added documents do not just sit alongside the existing corpus: they immediately enter its neighbourhood structure and can be retrieved as nearest neighbours of any prior chunk. In other words, ingestion is not file storage; it is *geometric admission*—each new chunk takes its place inside a single, evolving semantic manifold.

Analyst question: “Where in the existing corpus does this new document actually sit?”

3 Vector Similarity

Vector similarity is the geometric bridge between the embedded corpus of Section 2 and every retrieval or graph operation that follows. Two implementations of similarity coexist in the codebase: a streaming, index-backed similarity used for top- k retrieval inside Qdrant (`store.similarity_search` and the BFS exploration of Section 4), and a dense matrix form $W \in \mathbb{R}^{n \times n}$ computed in NumPy over a working subgraph (`singular.py`, `similarity_graph.py`). This section makes the two views explicit and shows how the dense form W becomes the unique input to the three downstream ranking strategies.

3.1 Step 1 — The Cosine Similarity Matrix (`singular.py`)

3.1.1 Main steps

1. **Embedding matrix.** Stack the n retrieved chunk embeddings into $\Phi \in \mathbb{R}^{n \times d}$, with row i equal to $\phi_i^\top$. Each ϕ_i is (approximately) ℓ_2 -unit, by the normalisation convention of Section 2.3.
2. **Gram matrix.** Compute the symmetric, positive-semidefinite Gram matrix

$$W^{\text{raw}} = \Phi \Phi^\top \in \mathbb{R}^{n \times n}, \quad W_{ij}^{\text{raw}} = \phi_i^\top \phi_j \approx \cos \angle(\phi_i, \phi_j).$$

3. **Threshold sparsification.** Apply the single similarity threshold $\tau = \text{SIMILARITY_THRESHOLD}$ (default 0.65):

$$W_{ij} = \begin{cases} W_{ij}^{\text{raw}} & \text{if } W_{ij}^{\text{raw}} \geq \tau \text{ and } i \neq j, \\ 0 & \text{otherwise.} \end{cases}$$

The diagonal is forced to zero so that W is the adjacency weight of a loop-free undirected graph.

3.1.2 What it produces

A symmetric matrix $W \in [0, 1]^{n \times n}$ with $W_{ii} = 0$ and edge density controlled by τ . W is the single mathematical object consumed by all three strategies of Section 4: spectral (Singular), geodesic (Hinge), and theta-based (Frontier).

Mathematical Grounding

Cosine geometry. For ℓ_2 -unit vectors, $\phi_i^\top \phi_j = \cos \angle(\phi_i, \phi_j) \in [-1, 1]$, and the squared Euclidean distance is $\|\phi_i - \phi_j\|_2^2 = 2(1 - \phi_i^\top \phi_j)$. On the unit sphere, cosine similarity, maximum inner product and nearest-neighbour rankings therefore coincide; the choice between them is purely numerical, not semantic.

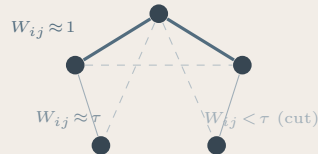
Spectral structure of W^{raw} . $W^{\text{raw}} = \Phi \Phi^\top$ is positive semidefinite with rank at most d . Its nonzero eigenvalues are exactly the squared singular values of Φ , and its eigenvectors span the principal axes of the chunk cloud in $\mathbb{R}^d$. The thresholded matrix W is no longer guaranteed PSD: thresholding breaks the kernel structure and is what justifies treating W as a *graph adjacency*, not as a kernel matrix, for the downstream Laplacian analysis.

Threshold as a calibration knob. The single scalar τ governs (i) the edge set of the BFS-induced graph; (ii) the connectivity of W ; (iii) the interdependence graph $H = \mathbf{1}_{W \geq \tau}$ that feeds the θ relaxation (Section 4.3). Lowering τ produces a denser graph with smoother spectrum; raising τ fragments the graph and inflates the multiplicity of $\lambda = 0$. A principled calibration of τ from the empirical similarity distribution is part of the validation programme of Section 6.

Why zero the diagonal? Spectral methods require an adjacency, not a kernel: with $W_{ii} = \|\phi_i\|^2 = 1$ on the diagonal, the Laplacian $L = D - W$ —where $D = \text{diag}(d)$ is the diagonal degree matrix with entries $d_i = \sum_j W_{ij}$ (formally introduced in Section 4.2)—would be biased by self-loops, and the normalised Laplacian’s null space would no longer encode connected components cleanly.

Semantic Interpretation (heuristic)

(Heuristic) W is the corpus seen as a weighted conversation graph.



Every off-diagonal entry W_{ij} says: “chunk i and chunk j talk about the same thing, with confidence W_{ij} ”. Strong edges trace explicit topical overlap; weak edges trace shared vocabulary or loose thematic kinship; absent edges signal semantic independence.

The threshold τ is the *editorial line* of the system: it decides which weak associations are worth keeping as evidence of connection. A low τ encourages permissive thematic linking (good for exploratory work, dangerous for inference); a high τ enforces a strict notion of relatedness (good for contrastive analysis, but risks fragmentation). Every downstream claim about poles, hinges and frontiers is, in the end, a claim relative to this single number.

Analyst question: “Above what level of similarity do I accept that two passages are speaking about the same thing?”

3.2 Step 2 — Top- k Retrieval (`store.similarity_search`)

3.2.1 Main steps

1. **Query encoding.** Embed the prompt q as $\phi(q) = f_\theta(q) \in \mathbb{R}^d$.
2. **Index lookup.** `client.query_points(collection, query= $\phi(q)$ , limit= $k$ )` traverses

the HNSW index and returns the top- k points

$$\mathcal{N}_k(q) = \operatorname{argtop}_k\{\phi(q)^\top \phi_i : i = 1, \dots, N\},$$

where argtop_k denotes the set of indices achieving the k largest values of the expression (and, symmetrically, $\operatorname{argbottom}_k$ the k smallest).

3. **Payload projection.** Each returned hit is flattened to a plain dictionary `{text,filename,chunk_number,score}`, ready for display or insertion into a RAG prompt.

3.2.2 Two retrieval regimes

The same cosine similarity underlies two complementary modes:

- *Top- k retrieval* (`similarity_search`, used in `pipelines/rag.py`): a single shot, k results, no graph structure — classical dense retrieval.
- *BFS exploration* (`exploration.py`, Section 4.1): a sequence of top- k lookups, each query being a stored point ID rather than a fresh embedding, accumulating a working subgraph.

The two regimes are not redundant: top- k retrieval optimises *relevance to the prompt*, BFS exploration optimises *coverage of the prompt's neighbourhood*. The graph-ranking strategies of Section 4 operate on the second.

Mathematical Grounding

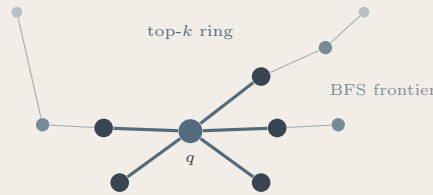
Top- k as a maximum-inner-product problem. For ℓ_2 -unit embeddings, $\operatorname{argtop}_k\{\phi(q)^\top \phi_i\} = \operatorname{argtop}_k\{\cos \angle(\phi(q), \phi_i)\} = \operatorname{argbottom}_k\{\|\phi(q) - \phi_i\|_2\}$; all three formulations select the same set.

Approximate vs. exact. HNSW returns approximate top- k with high recall but no exact guarantee [1, 2]. For analyst-scale collections ($N \lesssim 10^6$) the approximation error is empirically negligible relative to embedding noise; the caveat is the same asymmetry as in Section 4.1.

Retrieval vs. exploration. Top- k retrieval answers “which chunks are closest to the prompt?”; BFS exploration answers “which chunks are reachable from the prompt through the similarity graph?”. The first is the right primitive for direct question-answering RAG; the second is the right primitive for the geometric-attention rankings of Section 4.

Semantic Interpretation (heuristic)

(Heuristic) Two modes, two analyst questions.



The inner ring around the query q is the realm of top- k retrieval: chunks the prompt directly asks for. The outer layers are the realm of BFS exploration: chunks that the prompt’s neighbourhood pulls in, even though q itself never mentions them.

The first is what a user expects from a search box; the second is what an analyst needs to detect ambient context, contested concepts and blind spots. The same cosine similarity powers both—only the traversal policy differs.

Analyst question: “Do I want what the prompt asked for, or what the prompt’s world looks like?”

4 Graph Exploration

4.1 Step 0 — Building the Subgraph (exploration.py)

4.1.1 What BFS means here

Breadth-First Search (BFS) is a graph traversal strategy that explores nodes layer by layer, always processing the closest unvisited neighbours before moving further away. Starting from a seed set at distance 0, it visits all their neighbours (distance 1), then those neighbours’ neighbours (distance 2), and so on, until a stopping criterion is met.

In the Eigenmind - Singularities context, the “graph” traversed is the Qdrant approximate-nearest-neighbour (ANN) index: edges connect each stored vector to its k closest vectors in embedding space. BFS over this index collects a working subgraph by following ANN links outward from the query prompt.

Caveat: approximation and asymmetry.. The resulting subgraph carries three structural limitations [1, 2]:

- *ANN approximation*: Qdrant’s HNSW index does not return exact nearest neighbours.
- *Asymmetry*: ANN graphs are generally directed; u may be among v ’s k nearest neighbours without the converse. The subsequent cosine matrix $W = \Phi\Phi^\top$ restores symmetry.
- *Embedding dependence*: semantic connectivity is only as good as the underlying encoder [3].

BFS guarantees connectivity within the *induced ANN retrieval graph*, whose topology *approximates* local neighbourhood structure in embedding space.

4.1.2 Main steps

1. **Seed.** Encode the prompt $q \mapsto \phi(q) \in \mathbb{R}^d$. Retrieve the K_0 closest stored vectors (with K_0 the per-node fan-out, i.e. `NEIGHBORS_TO_FETCH` in `config.py`); initialise BFS queue $\mathcal{Q}$ and visited set $\mathcal{V}$.

2. **BFS expansion.** While $|\mathcal{V}| < N_{\max}$ (with $N_{\max}$ the target subgraph size, i.e. `MAX_CHUNKS_FOR_CONTEXT`) and $\mathcal{Q} \neq \emptyset$: dequeue v , retrieve its K_0 nearest neighbours excluding $\mathcal{V}$, enqueue new IDs.
3. **Batch retrieval.** Fetch all vectors $\{\phi_i\}_{i \in \mathcal{V}}$ and payloads in one batched call.
4. **Similarity matrix.**

$$W = \Phi \Phi^\top, \quad W_{ij} \leftarrow 0 \text{ if } W_{ij} < \tau, \quad W_{ii} = 0.$$

4.1.3 What it produces

A symmetric matrix $W \in [0, 1]^{n \times n}$ encoding pairwise semantic affinity over the retrieved subgraph.

4.2 Strategy 1 — Thematic Diversity (`singular.py`)

4.2.1 Main steps

1. **Normalised Laplacian.** $L_{\text{sym}} = I - D^{-1/2} W D^{-1/2}$, $D = \text{diag}(d)$, $d_i = \sum_j W_{ij}$. Isolated nodes ($d_i = 0$) get zero $D^{-1/2}$ entry.
2. **Full eigendecomposition.** Compute $0 = \lambda_0 \leq \lambda_1 \leq \dots \leq \lambda_{n-1}$ and eigenvectors of L_{sym} (symmetric solver, $O(n^3)$, practical for $n \lesssim 1,000$).
3. **Low-frequency mode selection.** The code counts near-zero eigenvalues: $k = |\{j : \lambda_j < 10^{-3}\}|$. Formally, the multiplicity of $\lambda = 0$ equals the number of connected components of W . In practice, BFS-retrieved graphs are almost always connected ($k = 1$); meaningful thematic structure lives in the *small nonzero* eigenvalues and their spectral gaps [6].
4. **Pole extraction.** For each selected eigenvector u_j ($j = 1, \dots, k$):

$$i_j^+ = \arg \max_i u_j(i), \quad i_j^- = \arg \min_i u_j(i).$$

The pair $\{i_j^+, i_j^-\}$ are the **Thematic Representatives** of mode j .

Mathematical Grounding

Mathematical grounding: Fiedler vector [5]. For the normalised Laplacian L_{sym} , the low-frequency eigenvectors minimise a *normalised Dirichlet energy*:

$$\mathcal{E}(u) = \frac{u^\top L_{\text{sym}} u}{\|u\|_2^2} = \frac{\sum_{i,j} W_{ij} (u_i - u_j)^2 / \sqrt{d_i d_j}}{\sum_i u_i^2},$$

and encode the most slowly varying functions on the degree-weighted graph. The first nontrivial minimiser u_1 (Fiedler vector) has its extrema at the *spectral antipodes*—the two nodes most separated along the dominant axis of thematic variation.

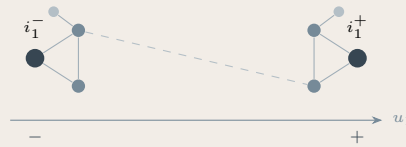
Formal theorem: Cheeger inequality [4]. $\lambda_1/2 \leq h(G) \leq \sqrt{2\lambda_1}$, where $h(G) = \min_S \frac{\text{cut}(S, \bar{S})}{\min(\text{vol}(S), \text{vol}(\bar{S}))}$ is the Cheeger (isoperimetric) constant of the weighted graph G induced by W . A small first nonzero eigenvalue λ_1 signals a weakly connected cut or metastable thematic separation—not bipartiteness. Bipartite-like structure is instead associated with eigenvalues near 2 in the normalised Laplacian spectrum.

Mathematical intuition: spectral gaps and diffusion modes. Higher eigenvectors reveal secondary axes [6, 7]; a pronounced eigengap between λ_j and λ_{j+1} signals a natural j -way thematic partition.

Important precision. Selecting eigenvector extrema identifies *spectral antipodes*, not “representative themes” in a formal sense. Whether these nodes are semantically representative depends on embedding quality and corpus structure, and should be validated by domain experts.

Semantic Interpretation (heuristic)

(Heuristic) Thematic Representatives anchor the axes of thematic diversity.



In a de-dollarisation corpus the poles might be:

- i_1^- : a central-bank analysis of U.S. Treasury market depth [21],
- i_1^+ : a passage on BRICS cross-border CBDC infrastructure [20].

In an inflation corpus, the poles might oppose supply-chain fragmentation [22] against monetary normalisation [23].

Analyst question: “What are the dominant axes of thematic variation in this corpus?”

4.3 Strategy 2 — Hinge Connectors (connectivity.py)

4.3.1 Main steps

1. **Length transform.** $\ell_{ij} = -\log W_{ij} \in [0, +\infty)$; $\ell_{ij} = +\infty$ if $W_{ij} = 0$.
2. **All-pairs geodesic distances.**

$$d(u, v) = \min_{\gamma: u \rightarrow v} \sum_{(i,j) \in \gamma} \ell_{ij}.$$

Disconnected pairs: penalty $10 \cdot \max_{\text{finite}} d$. Complexity $O(n^2 \log n + nm)$ where $m = |\{(i, j) : W_{ij} > 0\}|$ is the number of edges of W ; practical for $n \lesssim 500$.

3. **Peripheral source and geodesic field.** $v^* = \arg \max_v \sum_j d(v, j)$; $x_i^{\text{raw}} = d(v^*, i)$;

$$\bar{x} = \frac{\sum_i d_i x_i^{\text{raw}}}{\sum_i d_i}, \quad x_i = \frac{x_i^{\text{raw}} - \bar{x}}{\max_i |x_i^{\text{raw}} - \bar{x}|} \in [-1, 1].$$

4. **Pole definition.** $P_+ = \{i : x_i \geq q_{0.9}\}$, $P_- = \{i : x_i \leq q_{0.1}\}$ (fallback 80/20).

5. **Hinge score.**

$$S_{\pm}(i) = \sum_{j \in P_{\pm}} W_{ij}, \quad B(i) = \min(S_+, S_-), \quad H(i) = B(i) \cdot (1 - |x_i|).$$

Mathematical Grounding

Formal motivation: log-similarity geodesics [10]. A path of length $\sum (-\log W_{ij})$ is minimised when it maximises $\prod W_{ij}$ —the most coherent semantic route.

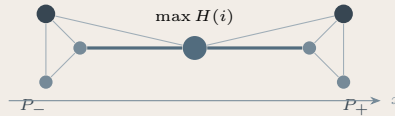
Mathematical intuition: geodesic field as heuristic proxy. A natural ℓ_{∞} analogue of the Fiedler problem would seek the smoothest function saturating the hypercube $\|x\|_{\infty} = 1$ with fixed pole constraints (i.e. $x_{i+} = +1$, $x_{i-} = -1$). However, the unconstrained problem $\arg \min_{\|x\|_{\infty} \leq 1, x \perp_w \mathbf{1}} x^{\top} L_{\text{sym}} x$ —where $x \perp_w \mathbf{1}$ denotes degree-weighted orthogonality to the constant vector, i.e. $\sum_i d_i x_i = 0$ —has the trivial minimiser $x = 0$; a non-trivial solution requires either a saturating norm equality or explicit boundary conditions. The implemented normalised geodesic field $x \in [-1, 1]^n$ should therefore be understood as a *heuristic proxy*: it lies on the ℓ_{∞} sphere by construction and is smooth in the geodesic sense, but it is not the solution of a well-posed ℓ_{∞} variational problem. Hinge connectors are nodes near the zero level set of this field.

Hinge functional. $B(i) = \min(S_+, S_-)$ rewards balanced dual affinity; $(1 - |x_i|)$ penalises nodes too close to either extreme [11].

Known limitations. Sensitivity to sparsification threshold τ , semantic hubness, and embedding anisotropy [10].

Semantic Interpretation (heuristic)

(Heuristic) Connectors are plausible conceptual bridges.



In a U.S.–China competition corpus [24], hinge chunks may be passages on *semiconductor supply chains* or *rare-earth logistics*—appearing in both containment and interdependence arguments.

In a monetary policy corpus, hinge chunks may discuss *labour market slack* [23]—relevant to both hawks and doves.

Analyst question: “Which concepts bridge the two thematic extremes?”

4.4 Strategy 3 — Frontier Themes (theta.py)

4.4.1 Main steps

1. **Interdependence graph.** $H_{ij} = \mathbf{1}_{W_{ij} \geq \tau}$, $H_{ii} = 0$. (W symmetric $\Rightarrow H$ symmetric.)
Using the same threshold τ as the similarity graph means that any edge present in

W also constitutes an interdependence: the two chunks are informationally redundant whenever they are directly connected.

2. Subgradient approximation of the Lovász dual.

$$A^{(k)} = \text{diag}(\lambda^{(k)}) + \frac{1}{4}(\mu^{(k)} + (\mu^{(k)})^\top) \odot H, \quad x^{(k)} = \frac{1}{2}(A^{(k)})^{-1}(\mathbf{1} + \lambda^{(k)}),$$

where $\odot$ denotes the Hadamard (element-wise) matrix product. Step $t_k = t_0/\sqrt{k}$; best dual bound θ^* retained.

3. Frontier matrix and scores.

$$\mathcal{F} = \frac{(A^*)^{-1}}{\text{tr}[(A^*)^{-1}]} \succeq 0, \quad \mathcal{F} \approx YY^\top, \quad \text{FS}(i) = \mathcal{F}_{ii} = \|y_i\|^2.$$

4. Farthest-point selection. $\hat{y}_i = y_i/\|y_i\|$; seed = highest SVD leverage; then $i^{(t+1)} = \arg \max_i \min_{j \in S^{(t)}} (1 - \hat{y}_i^\top \hat{y}_j)$.

Mathematical Grounding

Primal problem. $\alpha(H) = \max_{x \in \{0,1\}^n} \{\mathbf{1}^\top x \mid x_i x_j = 0 \ \forall (i,j) \in E(H)\}$ (with $E(H) = \{(i,j) : H_{ij} = 1\}$ the edge set of H and $\alpha(H)$ the independence number), with binary constraints $(x_i - x_i^2 = 0)$ and independence constraints.

Lagrangian (Lemaréchal–Oustry [15]).

$$\mathcal{L}(x, \lambda, \mu) = \mathbf{1}^\top x + \sum_i \lambda_i (x_i - x_i^2) - \frac{1}{2} \sum_{(i,j) \in E(H)} \mu_{ij} x_i x_j = (\mathbf{1} + \lambda)^\top x - x^\top A(\lambda, \mu) x.$$

Formal theorem: dual problem and Lovász theta [13].

$$g(\lambda, \mu) = \frac{1}{4}(\mathbf{1} + \lambda)^\top A^{-1}(\mathbf{1} + \lambda), \quad \min_{\lambda, \mu \geq 0, A \succ 0} g = \vartheta(H) \geq \alpha(H); \quad \alpha(H) \leq \vartheta(H) \leq \chi(\bar{H}),$$

where $\bar{H}$ is the complement graph of H ($\bar{H}_{ij} = 1 - H_{ij}$ off-diagonal) and $\chi(\cdot)$ denotes the chromatic number. For the *canonical SDP formulation*, strong duality holds under standard SDP regularity assumptions [14]. The implemented subgradient scheme should be understood as an *approximation* to this relaxation: it is a Lovász-theta-inspired Lagrangian/SDP relaxation proxy for semantic independence, inheriting its motivation from the theta relaxation, but requiring empirical validation before interpreting the resulting frontier scores as faithful ϑ -leverage scores.

Mathematical Grounding

The dual function $g(\lambda, \mu)$ is *convex* in (λ, μ) : it is a pointwise maximum of functions affine in (λ, μ) (for each fixed x , the Lagrangian $\mathcal{L}(x, \lambda, \mu)$ is affine in (λ, μ) , and the supremum of any family of affine functions is convex). Its *subdifferential* at (λ, μ) is the set of all subgradients—vectors (s_λ, s_μ) satisfying the global supporting-hyperplane inequality:

$$g(\lambda', \mu') \geq g(\lambda, \mu) + s_\lambda^\top (\lambda' - \lambda) + \langle s_\mu, \mu' - \mu \rangle \quad \forall (\lambda', \mu').$$

By *Danskin's theorem* [18], for a parametric maximum $g(\theta) = \max_x \mathcal{L}(x, \theta)$, the subdifferential is the convex hull of gradient vectors evaluated at all maximisers:

$$\partial_{(\lambda, \mu)} g = \text{conv} \left\{ (\nabla_\lambda \mathcal{L}(x^*, \lambda, \mu), \nabla_\mu \mathcal{L}(x^*, \lambda, \mu)) : x^* \in X^*(\lambda, \mu) \right\},$$

where $X^*(\lambda, \mu) = \arg \max_x \mathcal{L}(x, \lambda, \mu)$. When $A(\lambda, \mu) \succ 0$ the maximiser is unique ($x^* = \frac{1}{2} A^{-1}(\mathbf{1} + \lambda)$), the convex hull collapses to a singleton, and g is *differentiable* at (λ, μ) with gradient equal to the subgradient.

Computing the subgradient. Differentiating the Lagrangian with respect to each multiplier family and evaluating at x^* gives the two blocks of the gradient:

$$\begin{aligned} \left. \frac{\partial \mathcal{L}}{\partial \lambda_i} \right|_{x^*} &= x_i^* - (x_i^*)^2 & \Rightarrow \quad \nabla_\lambda g &= x^* - (x^*)^{\circ 2}, \\ \left. \frac{\partial \mathcal{L}}{\partial \mu_{ij}} \right|_{x^*} &= -\frac{1}{2} H_{ij} x_i^* x_j^* & \Rightarrow \quad \nabla_\mu g &= -\frac{1}{2} x^*(x^*)^\top \odot H, \end{aligned}$$

where $(\cdot)^{\circ 2}$ denotes the element-wise square.

The minimisation of g over the feasible domain $\{\lambda \in \mathbb{R}^n, \mu \geq 0, A \succ 0\}$ therefore proceeds by:

- *Descent on λ* (unconstrained): $\lambda^{(k+1)} = \lambda^{(k)} - t_k(x^{*(k)} - (x^{*(k)})^{\circ 2})$;
- *Projected descent on μ* ($\mu \geq 0$ constraint): $\mu^{(k+1)} = \max(0, \mu^{(k)} + \frac{t_k}{2} x^{*(k)}(x^{*(k)})^\top \odot H)$.

Since $\nabla_\mu g \leq 0$ (element-wise, for feasible x^* in $[0, 1]^n$), projecting the descent step is equivalent to *increasing* μ by $\frac{t_k}{2} x^*(x^*)^\top \odot H$ —exactly what the code performs (with the factor $\frac{1}{2}$ absorbed into t_0). The step size $t_k = t_0/\sqrt{k}$ is the classical *diminishing step rule* introduced by Polyak [16] and further analysed by Nesterov [17], which guarantees convergence of the running best $\theta^* = \min_k g(\lambda^{(k)}, \mu^{(k)})$ to $\vartheta(H)$ as $k \rightarrow \infty$ when g is Lipschitz—though the practical finite-iteration approximation carries no formal certificate [18].

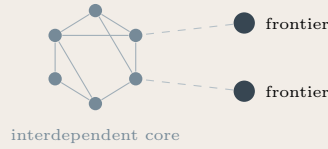
Approximation status. The subgradient iteration provides *no formal convergence certificate* within finite iterations [18]; frontier scores are proxies, not exact SDP optima.

Shannon capacity analogy (not a formal equivalence). Lovász introduced ϑ to bound the zero-error channel capacity $C(G) = \sup_k \alpha(G^{\boxtimes k})^{1/k}$ [19, 13], where $G^{\boxtimes k}$ is the k -fold strong product of G with itself ($\boxtimes$ denotes the strong graph product: adjacency in $G_1 \boxtimes G_2$ holds whenever the corresponding components are equal or adjacent in their respective graphs). Under the analogy that interdependent chunks ($H_{ij} = 1$) are “confusable symbols”, frontier themes correspond to a maximum-capacity non-confusable code of the corpus. This framing requires verification that τ correctly models the channel structure.

Frontier matrix. $\mathcal{F}_{ij} \approx 0$ for interdependent pairs; $\mathcal{F}_{ii} = \|y_i\|^2$ measures each chunk’s independent information proxy.

Semantic Interpretation (heuristic)

(Heuristic) Frontier Themes are plausible weak signals and blind spots.

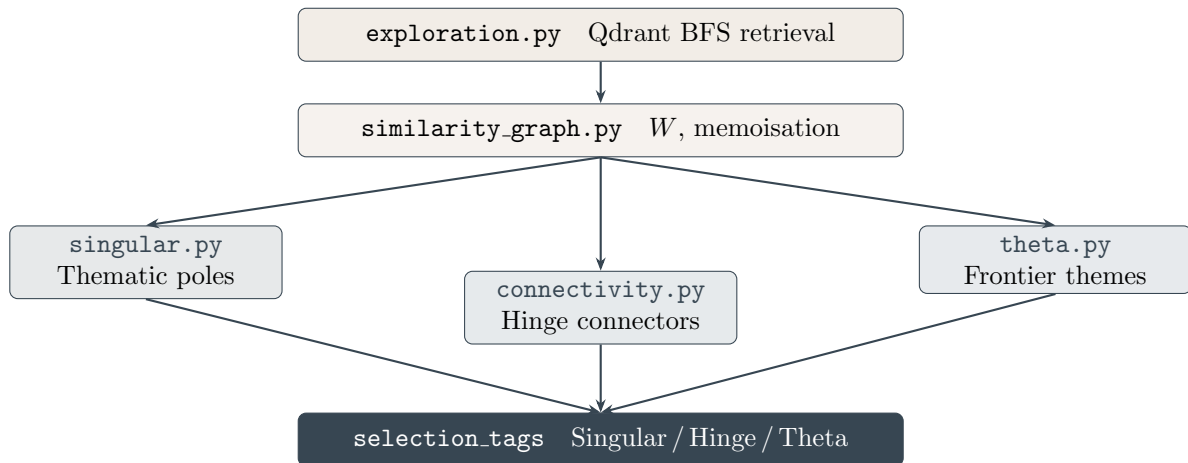


In a Middle-East corpus, frontier chunks may be passages on *water diplomacy* [25] or *Arctic shipping implications* [26]—present but not amplified by the dominant discourse. In global macro research, frontier chunks may surface *collateral scarcity* [27] or *pension fund reallocation* [28].

Analyst question: “Which chunks carry unique, non-redundant information?”

4.5 The Pipeline in One Picture

The first two stages are sequential: `exploration.py` retrieves the working subgraph from Qdrant, and `similarity_graph.py` builds and memoises W . The three ranking strategies then run in *parallel* from the same W , and their outputs are merged by `selection_tags`.



The `SimilarityGraph` class holds W and lazily dispatches to the three strategy modules, caching intermediate results.

4.6 Practical Scope, Assumptions, and Failure Modes

4.6.1 Intended operating regime

Eigenmind - Singularities operates on analyst-scale, locally retrieved semantic subgraphs.

Operation	Complexity	Practical limit
Full eigendecomposition	$O(n^3)$	$n \lesssim 1,000$
All-pairs Dijkstra	$O(n^2 \log n)$	$n \lesssim 500$
Subgradient dual (T iters)	$O(Tn^2)$	$n \lesssim 300\text{--}500$

4.6.2 Formal assumptions

1. *Embedding smoothness*: semantically similar documents have high cosine similarity.
2. *Graph connectivity*: the similarity graph W should have a small number of connected components.
3. *Threshold calibration*: the single threshold τ governs both the similarity graph W and the interdependence graph H ; it should be set with domain awareness.

4.6.3 Known failure modes

- **Embedding collapse**: out-of-domain encoders map distinct documents to nearby vectors; all three strategies degenerate.
- **Semantic hubness**: high-degree nodes dominate geodesic paths, biasing the hinge field.
- **Graph fragmentation**: too-aggressive τ disconnects the graph; Laplacian has multiple zero eigenvalues.
- **Adversarial/noisy corpora**: repetitive documents inflate H , saturating frontier scores.

4.6.4 Empirical validation (future work)

1. *Stability analysis*: sensitivity to prompt perturbations, edge removal, encoder substitution.
2. *Human evaluation*: expert assessment of thematic, hinge, and frontier labels.
3. *Baseline comparison*: PageRank, betweenness centrality, HDBSCAN, k -means, MMR [12], standard RAG.
4. *Quantitative novelty*: semantic coverage, redundancy reduction, expert novelty scoring.

4.7 Aggregation — The SimilarityGraph Class

4.7.1 Main steps

`selection_tags(top_k)` proceeds as follows:

1. `singular_chunks()` $\rightarrow$ tag "Singular".
2. `hinge_ranking()` (memoised x^*) $\rightarrow$ tag "Hinge" on top- k .
3. `theta_diversity(k)` (memoised Y) $\rightarrow$ tag "Theta" on top- k .

Spectrum, x^* , and Y are each computed at most once per instance.

4.7.2 The three-label output

Tag	What it means	Analyst question
Singular	Spectral antipode along a low-frequency mode	What are the dominant axes of thematic variation?
Hinge	Near the ℓ_∞ zero level set, connected to both poles	Which concepts bridge the two thematic extremes?
Theta	Low interdependence proxy; high frontier score $FS(i)$	Which chunks carry unique, non-redundant information?

4.7.3 What multi-label chunks reveal

- **Singular + Hinge:** a spectral antipode also connected to the opposite pole—a contested concept claimed by both sides.
- **Hinge + Theta:** a bridge concept that is also informationally self-contained—a potential coalition-enabling idea.
- **All three:** highest structural density; read first.

4.7.4 Design principle: named chunks, not latent vectors

Every tag is attached to a named, readable chunk. Analysts can read, verify, and annotate each tagged chunk—maintaining epistemic control over a mathematical pipeline.

4.8 What Is Formal, What Is Heuristic, What Is Empirical

The memo uses three levels of claim throughout. This section consolidates them in one place for scientific clarity.

Claim type	Examples in this memo	Status and caveat
Formal theorem	Zero multiplicity = connected components; Cheeger inequality; dual $g = \vartheta(H)$ at optimum; Lovász sandwich $\alpha \leq \vartheta \leq \chi(\bar{H})$	Proven. Applies to the mathematical objects as defined, under standard regularity assumptions (connected graph, $A \succ 0$, etc.).
Mathematical intuition	Low-frequency eigenvectors encode thematic variation; log-similarity geodesics follow high-affinity paths; geodesic field as ℓ_∞ -sphere proxy; spectral gaps signal thematic partitions	Grounded in established theory but <i>not</i> a formal guarantee in this specific corpus context. Depends on embedding quality, graph connectivity, and threshold calibration.
Heuristic semantic interpretation	Spectral poles as thematic archetypes; hinge connectors as negotiation bridges; frontier themes as weak signals; Shannon-capacity analogy for frontier selection	Working hypotheses only. Each claim requires corpus-specific empirical validation (expert review, stability analysis, baseline comparison).
Heuristic approximation	Subgradient scheme as proxy for $\vartheta(H)$; frontier scores $\text{FS}(i)$ as independence proxies	No formal convergence certificate within finite iterations. Results are ϑ -inspired, not exact SDP optima. Validation against exact solvers recommended on small held-out corpora.

The overall pipeline is scientifically defensible as a *geometric-semantic attention system*: each stage has a clear mathematical motivation, known limitations, and a well-defined failure mode. The gap between formal guarantees and deployed behaviour should be closed progressively through the empirical validation programme described in Section 6.

4.9 Summary

Module	Core operation	Math object	Semantic class
<code>singular.py</code>	Eigenvectors of L_{sym} ; spectral antipodes	Fiedler vector, Cheeger ineq.	Thematic Representatives
<code>connectivity.py</code>	Geodesic field; ℓ_∞ zero level set; hinge score	Log-similarity geodesics, ℓ_∞ Rayleigh quotient	Hinge Connectors
<code>theta.py</code>	Lemaréchal–Oustry Lagrangian dual; $\mathcal{F} = YY^\top$	Lovász ϑ , SDP relaxation (heuristic approx.)	Frontier Themes
<code>exploration.py</code>	BFS over approximate ANN index	Local neighbourhood (approximate)	Working subgraph
<code>similarity_graph.py</code>	Lazy memoised parallel dispatch	Union of three rankings	Singular / Hinge / Theta

The three strategies are geometrically complementary:

- **Singular** (*formal*): spectral antipodes on the ℓ_2 sphere, Cheeger-guaranteed dominant thematic axes.
- **Hinge** (*intuition*): zero level set of the ℓ_∞ Rayleigh minimiser, bridging thematic poles.
- **Theta** (*heuristic approx.*): Lemaréchal–Oustry dual of the Lovász SDP, surfacing informationally self-contained frontier themes.

Together they provide a geometry of analytical attention: poles orient the analyst in narrative space; [hinges](#) suggest where dialogue is possible; the frontier warns of what is being missed.

References

- [1] Malkov, Y. A. and Yashunin, D. A. (2018). Efficient and robust approximate nearest neighbor search using hierarchical navigable small world graphs. *IEEE TPAMI*, 42(4):824–836.
- [2] Johnson, J., Douze, M., and Jégou, H. (2019). Billion-scale similarity search with GPUs. *IEEE Trans. Big Data*, 7(3):535–547.
- [3] Reimers, N. and Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. *EMNLP-IJCNLP 2019*.
- [4] Chung, F. R. K. (1997). *Spectral Graph Theory*. AMS.
- [5] Fiedler, M. (1973). Algebraic connectivity of graphs. *Czechoslovak Mathematical Journal*, 23(2):298–305.
- [6] Luxburg, U. von (2007). A tutorial on spectral clustering. *Statistics and Computing*, 17(4):395–416.
- [7] Coifman, R. R. and Lafon, S. (2006). Diffusion maps. *Applied and Computational Harmonic Analysis*, 21(1):5–30.
- [8] Belkin, M. and Niyogi, P. (2003). Laplacian eigenmaps. *Neural Computation*, 15(6):1373–1396.
- [9] Shuman, D. I. et al. (2013). The emerging field of signal processing on graphs. *IEEE Signal Processing Magazine*, 30(3):83–98.
- [10] Tenenbaum, J. B., de Silva, V., and Langford, J. C. (2000). A global geometric framework for nonlinear dimensionality reduction. *Science*, 290(5500):2319–2323.
- [11] Newman, M. E. J. (2005). A measure of betweenness centrality based on random walks. *Social Networks*, 27(1):39–54.
- [12] Carbonell, J. and Goldstein, J. (1998). The use of MMR, diversity-based reranking. *SIGIR 1998*, pp. 335–336.
- [13] Lovász, L. (1979). On the Shannon capacity of a graph. *IEEE Trans. Inf. Theory*, 25(1):1–7.
- [14] Grötschel, M., Lovász, L., and Schrijver, A. (1988). *Geometric Algorithms and Combinatorial Optimization*. Springer.
- [15] Lemaréchal, C. and Oustry, F. (2001). SDP relaxations in combinatorial optimization from a Lagrangian viewpoint. In *Advances in Convex Analysis and Global Optimization*, pp. 119–134. Springer US.
- [16] Polyak, B. T. (1987). *Introduction to Optimization*. Optimization Software, New York.
- [17] Nesterov, Y. (2024). Primal subgradient methods with predefined step sizes. *Journal of Optimization Theory and Applications*, 203(3):2083–2115.
- [18] Boyd, S. and Vandenberghe, L. (2004). *Convex Optimization*. Cambridge University Press.
- [19] Shannon, C. E. (1956). The zero error capacity of a noisy channel. *IRE Trans. Inf. Theory*, 2(3):8–19.

- [20] BIS Innovation Hub (2022). *Project mBridge: Connecting Economies Through CBDC*. Bank for International Settlements.
- [21] Arslanalp, S., Eichengreen, B., and Simpson-Bell, C. (2022). *The Stealth Erosion of Dollar Dominance*. IMF Working Paper WP/22/58.
- [22] Cavallo, A. and Rigobon, R. (2016). The Billion Prices Project: Using online prices for measurement and research. *Journal of Economic Perspectives*, 30(2):151–178.
- [23] Yellen, J. L. (2014). *Labor Market Dynamics and Monetary Policy*. Jackson Hole speech.
- [24] Miller, C. (2022). *Chip War*. Scribner.
- [25] Lowi, M. R. (1993). *Water and Power*. Cambridge University Press.
- [26] PAME (2020). *Arctic Shipping Status Report*. Arctic Council.
- [27] Singh, M. and Stella, P. (2012). *Money and Collateral*. IMF Working Paper WP/12/95.
- [28] OECD (2023). *Pension Markets in Focus 2023*. OECD Publishing.